

10. Design of counter circuit using Vivado IP

(a) Design of 8-bit up-counter circuit with an increment value of 2

Aim

To simulate an 8-bit up-counter that increments by 2 on each clock cycle using Vivado IP and verify its correct operation through waveform analysis.

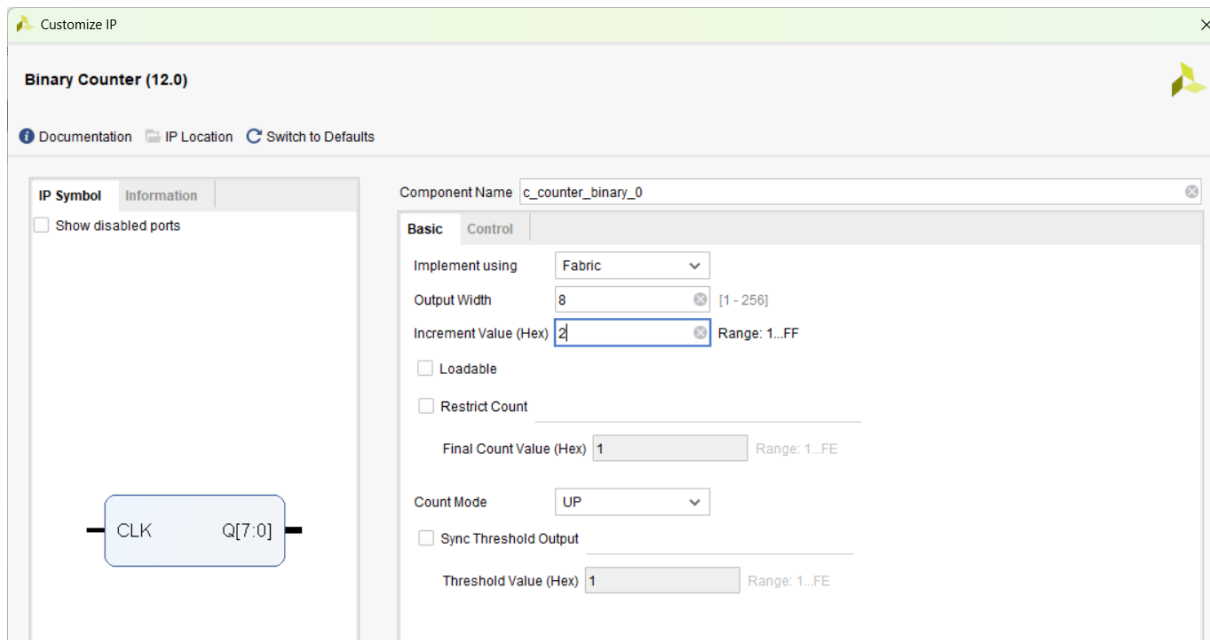
Software Required

- Xilinx Vivado Design Suite

Procedure

- 1. Create Vivado Project:**
 - Open Vivado and create a new project targeting the appropriate FPGA device.
- 2. Add Counter IP:**
 - Use Vivado's IP Catalog to add an up-counter or design a simple 8-bit counter with an increment of 2 using IP integrator or custom HDL.
- 3. Configure the Counter:**
 - Set the counter width to 8 bits.
 - Set increment value to 2 (connect a constant '2' to the increment input).
- 4. Write Testbench:**
 - Create a testbench that:
 - Provides a clock signal.
 - Resets the counter initially.
 - Monitors the output count over time.
- 5. Run Simulation:**
 - Launch Vivado Simulator.
 - Simulate for enough clock cycles to observe the counting sequence.
 - Observe the output waveform to verify the counter increments by 2 on each rising clock edge.
- 6. Analyze Results:**
 - Check that the counter output follows the sequence: 0, 2, 4, 6, ..., 254, 0 (rollover).

Counter IP configuration



Testbench program for Counter

```
`timescale 1ns / 1ps

module tb_clock_divider_by4;

reg clk;

reg rst;

wire [7:0] out1;

c_counter_binary_0 uut (

.CLK(clk),

.Q(out1)

);

always #5 clk = ~clk;

initial begin

    clk = 0;

    rst = 1;

    #20 rst = 0;
```

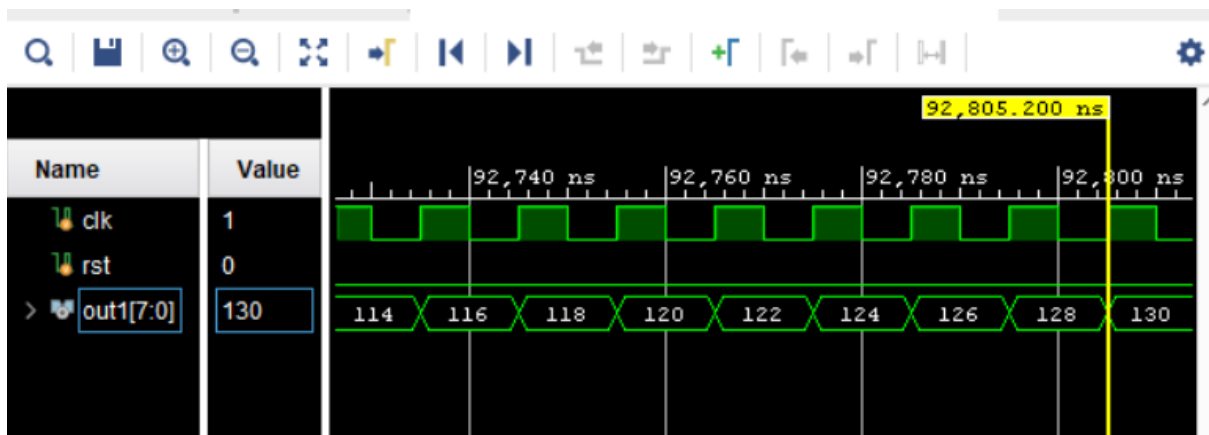
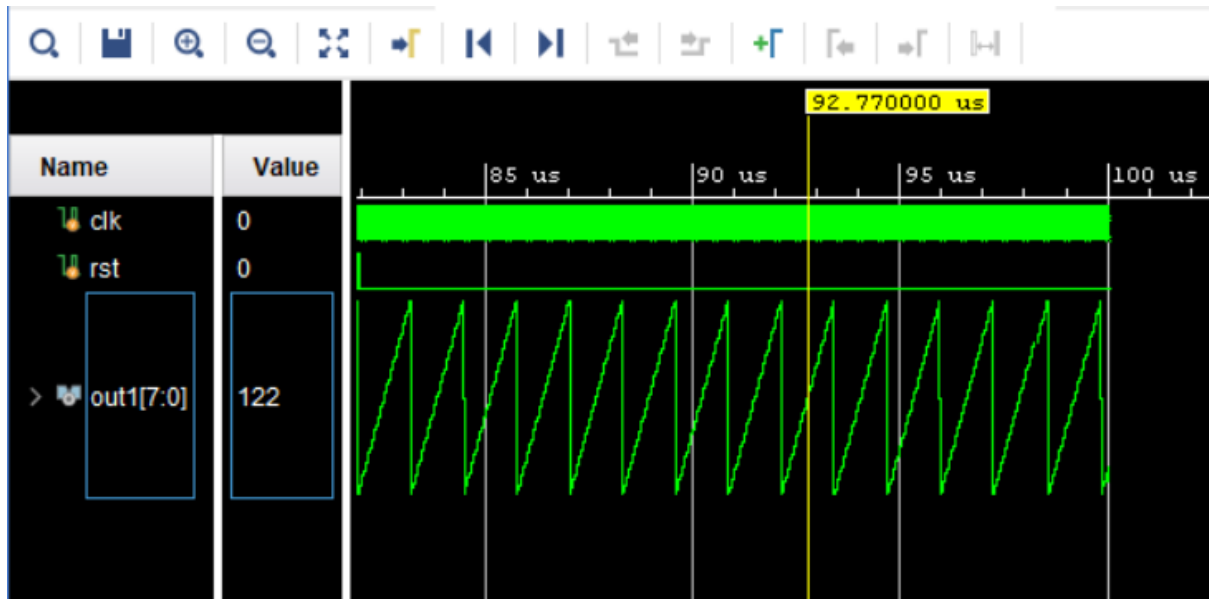
```
#100000;
```

```
$stop;
```

```
end
```

```
endmodule
```

Output



Result

- The simulation waveform showed the 8-bit counter incrementing by 2 on every clock cycle.
- The counter correctly rolled over after reaching the maximum value ($254 + 2 = 0$).
- The counter operation was verified to be accurate and stable.

(b) Design of 16-bit down-counter circuit with a decrement value of 1

Aim

To design and simulate a 16-bit down-counter using Vivado IP core and verify that the counter decreases by 1 on every active clock pulse.

Software required

- Xilinx Vivado Design Suite

Procedure

Step 1: Project Creation

Open Vivado and create a new RTL project. Enter the project name and location. Do not add sources at this stage. Select the appropriate FPGA device or board and complete the project setup.

Step 2: Adding Counter IP Core

Open the IP Catalog and search for Binary Counter. Select the counter IP core and customize it by setting the output width to 16 bits and selecting down-counter mode. Enable clock and reset options. Generate the IP core.

Step 3: HDL Wrapper and Top Module

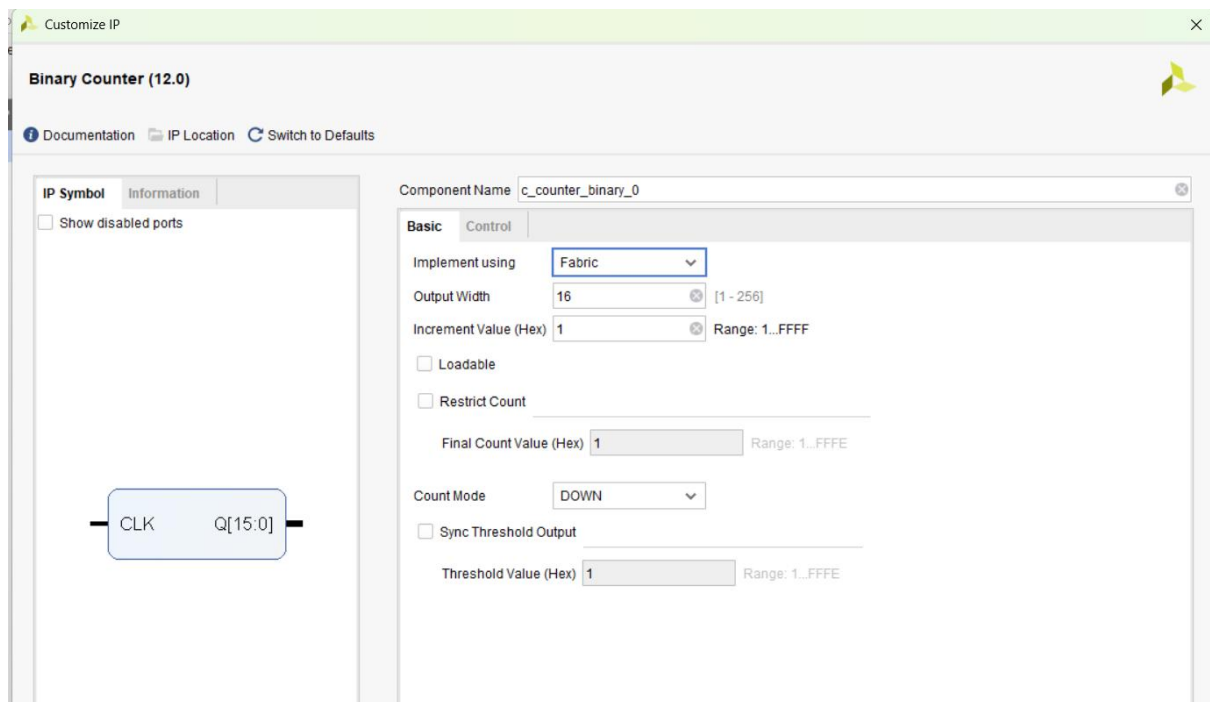
Create an HDL wrapper for the generated IP. Add a top module and connect the clock, reset, enable, and 16-bit output signals.

Step 4: Writing the Testbench Program

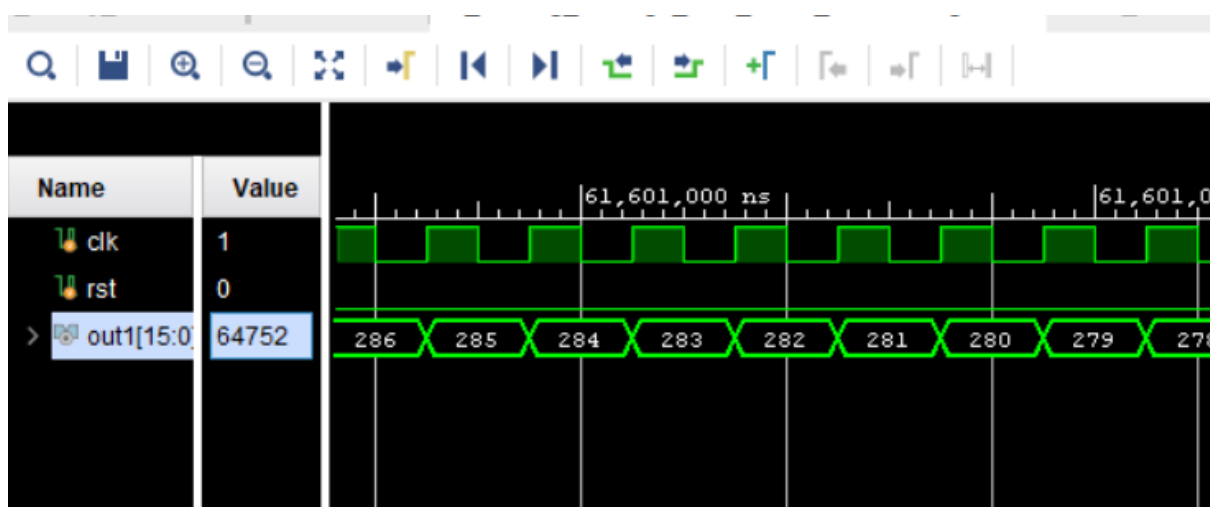
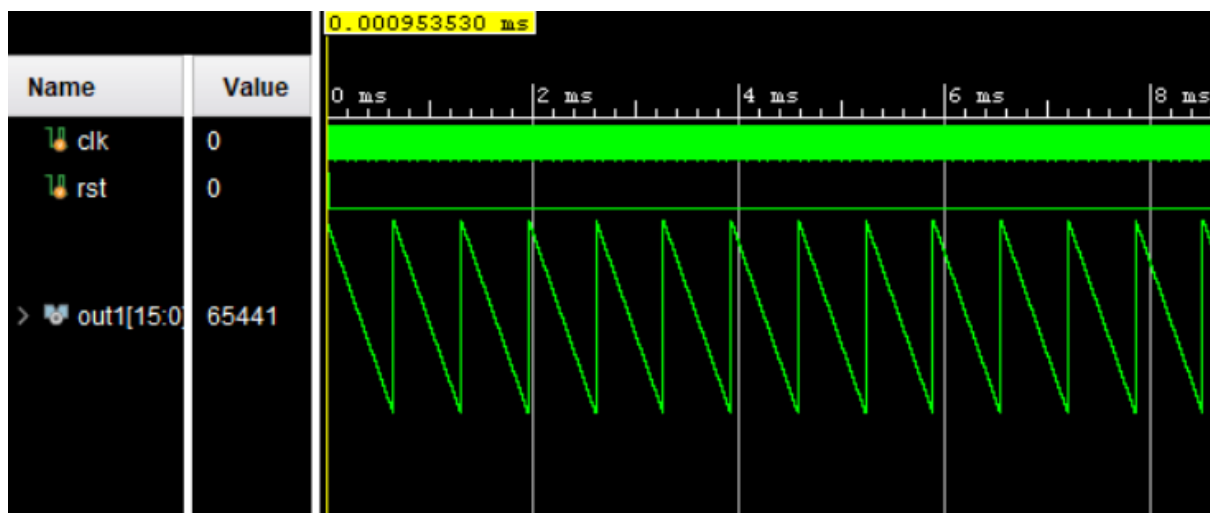
Add a new simulation source file and write a Verilog or VHDL testbench program. Generate a periodic clock signal. Apply reset initially. Provide enable signal and initialize the counter to a known value such as FFFFH.

Step 5: Running Simulation

Run behavioral simulation. Open the waveform window and observe the counter output. Verify that the count decreases by one for every clock pulse.



Output



Result

A 16-bit down-counter using Vivado IP was successfully designed and simulated. The waveform confirms that the counter decrements by 1 for each clock pulse, validating correct operation.